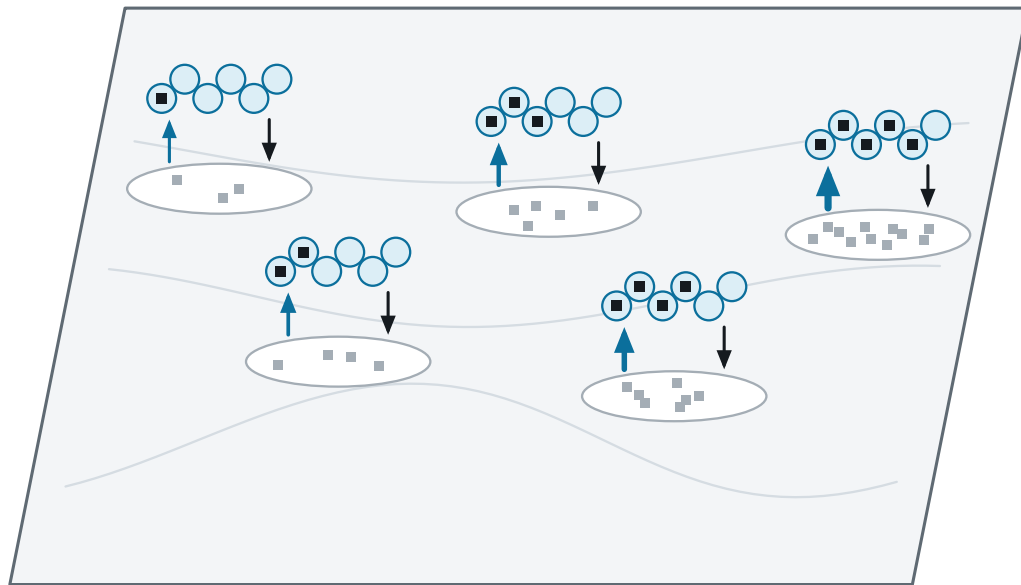


Gain and loss, place by place

no host movement, no effect on hosts: just the local environment

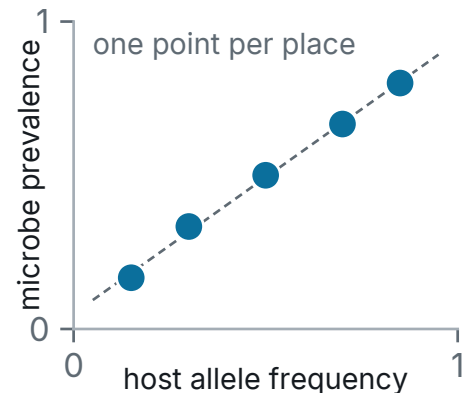
so the share of hosts carrying it settles at
gain ÷ (gain + loss), a different value in each place



↑ gain from the local environment (differs by place)

↓ loss (the same everywhere)

An association nobody selected for



if allele frequencies also differ among places, for any reason:

within any one place

allele and microbe: independent

all hosts pooled

associated ($r = 0.23$ here)

Through QME and the variance components  a first sketch

M_A  variation within places + differences among places

C_A  allele frequency and prevalence varying together among places

background, present before any selection on hosts; next: host movement